

Obfuscated Token Sequence Classification: Feature Engineering and Rigorous Validation for Human vs. Machine Text

Kaggle Team: Feature Forge — Gagan Eswar (27909) — Pratap (28367) — Aryan (26579)

Abstract

This project investigates text classification from obfuscated token sequences using sparse lexical, stylometric, and sequence-structural features. Lacking raw semantic information, we engineered a 150,027-feature Champion representation combining TF-IDF n-grams, stylometrics, generative Markov log-likelihoods, and Heaps' Law dynamics. After identifying data leakage in the initial cross-validation pipeline, we implemented fold-isolated multi-seed validation, establishing a project-specific empirical 94.38% hurdle for evaluating subsequent feature additions. The final experiments yielded two final ensemble configurations, both achieving 96.06% public leaderboard accuracy. The Phase B LightGBM + Logistic Regression ensemble using the 150,028-feature similarity-augmented Champion representation achieved 94.43% leak-free cross-validation accuracy, while the Phase 3 ensemble of standard and shallow LightGBM models achieved 94.46%. Thus, both configurations produced the same public leaderboard result, with only a small 0.03 percentage-point difference in cross-validation accuracy.

1 Problem and Data

The objective is to classify whether a document was authored by a human (Class A) or generated by a machine (Class B). The training data contains 10,536 labelled documents, and the test data contains 3,000 documents. The task uses an unusual representation where all text is anonymized into sequences of integer token IDs drawn from a vocabulary of $V = 18,438$. Documents vary in length from a few tokens to just under 2,000 tokens.

This representation removes access to raw textual context and semantic word meaning because the token mapping is not distributed. Consequently, classification was approached using statistical, token-frequency, local-sequence, and transition-level information. The dataset exhibits class imbalance (Class B $\approx 1.85 \times$ Class A), so CV splits were stratified and classifiers applied class weighting during training. While accuracy is the primary metric, minority-class detection is robust: Phase B achieved a 92.2% F1-score (90.3% precision, 94.3% recall) on Class A.

2 Feature Engineering and Model

To capture complementary signals from the integer sequences, four groups of features were engineered, resulting in a 150,027-dimensional representation:

- **TF-IDF Representation (150,000 features):** Sublinear-scaled TF-IDF n-grams (1–3 grams). Designed to capture token-specific frequency and local sequence patterns.
- **Stylometric Features (18 features):** Includes document length, unique-token counts, and token entropy. Designed to capture broader document-composition characteristics beyond individual n-gram frequencies.
- **Generative Markov Features (8 features):** Class-conditional sequence transition log-likelihoods. A dictionary-based mapping strategy was used for memory-efficient computation over the large vocabulary. This provides a compact summary of class-conditional transition behavior.
- **Heaps' Law (1 feature):** A vocabulary-growth exponent, capturing the rate of vocabulary growth with document length.

LightGBM formed the main base learner because the experimental pipeline could combine the sparse TF-IDF representation with dense engineered features while allowing nonlinear interactions among heterogeneous feature groups. Logistic Regression and a second shallow LightGBM were used in the two ensemble variants.

3 Validation and Leakage

Early modeling established a TF-IDF + stylometric baseline of 89.82% CV. Adding generative features substantially improved local performance to over 94%. However, investigation identified data leakage: TF-IDF vocabulary construction and generative feature fitting had been performed globally before fold splitting. Therefore, validation folds improperly influenced feature construction, causing optimistic estimates.

The pipeline was corrected. Every feature transformation was fitted exclusively on the training fold and then applied to the validation fold, and subsequent evaluation used this leak-free protocol.

Using five-fold stratified cross-validation across five random seeds (42, 100, 2026, 777, 12345), the corrected baseline yielded accuracies of 94.02%, 94.11%, 94.15%, 94.21%, and 94.31% (Mean = 94.16%). To consistently assess subsequent additions, a project-specific empirical Mean + 2σ hurdle of 94.38% was established (Figure 1). This empirical threshold is not a formal statistical significance test and is not a guarantee of generalization.

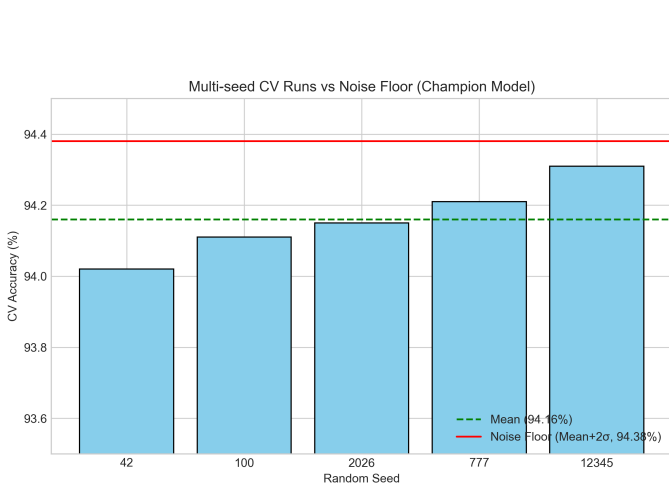


Figure 1: Multi-seed cross-validation results for the baseline LightGBM. The mean accuracy is 94.16%, with the empirical Mean + 2σ hurdle at 94.38%.

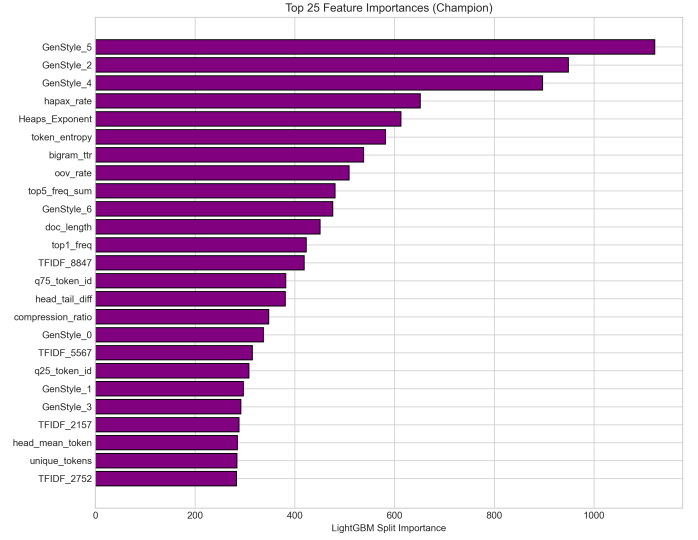


Figure 2: Top LightGBM feature importances, illustrating the relative importance assigned to Generative Markov, Heaps' Law, and TF-IDF features by the fitted model.

4 Experimental Comparison

Several alternative hypotheses were evaluated using the leak-free protocol and empirical threshold (Table 1).

Shape Statistics: We hypothesized that token-shape or statistical characteristics might contain information not captured by token identity and n-gram frequency. However, this addition yielded a leak-free CV of 94.12%, which did not exceed the 94.38% empirical hurdle. The feature family was therefore not retained.

Pseudo-labeling: To test semi-supervised generalization, Phase 4 incorporated pseudo-labeling using high-confidence test predictions. Under the leak-free evaluation protocol, Phase 4 achieved 94.58% CV accuracy and an AUC of 98.48%. This CV accuracy exceeded the project's 94.38% empirical hurdle by 0.20 percentage points. The experiment improved internal leak-free CV accuracy, but this gain did not transfer to the public leaderboard: its 95.53% LB score was below the Original Champion's 95.66%. This illustrates that higher internal validation does not necessarily translate to higher public leaderboard performance. One explanation is that pseudo-labeling introduces errors through model-generated labels, but current experiments do not isolate this mechanism.

Similarity-Augmented Champion: To investigate near-duplicate sequences, we added one dense feature capturing the maximum TF-IDF cosine similarity of each document to the training set. This configuration achieved 94.38% leak-free CV accuracy and 95.93% public leaderboard accuracy.

Phase B Stacking: A Phase B stacking ensemble was constructed using base models and a Logistic Regression meta-learner over out-of-fold predictions from the similarity-augmented 150,028-feature representation. This configuration achieved a leak-free CV of 94.43% and a public leaderboard score of 96.06%.

Phase 3 Ensemble: A Phase 3 ensemble combined two LightGBMs (one standard, one shallow) trained on the 150,028-feature representation. This configuration achieved a leak-free CV of 94.46% and a public leaderboard score of 96.06%.

5 Discussion and Limitations

TF-IDF n-grams preserve token identity and local sequence patterns. Markov likelihoods summarize class-conditional transition behavior. Heaps' Law captures vocabulary-growth behavior. Stylometrics summarize document-level composition. The similarity feature captures proximity to training documents. LightGBM combines these heterogeneous features and nonlinear interactions. The two final configurations (Phase B and Phase

Table 1: Comparison of major experimental and final modeling configurations.

Model	Representation / Method	CV Acc.	LB Acc.
Original Champion	150,027 features	94.16%	95.66%
Champion + Shape Statistics	150,027 + shape features	94.12%	—
Phase 4 Pseudo-labeling	150,027 + pseudo-labeling	94.58%	95.53%
Similarity-Augmented Champion	150,028 features	94.38%	95.93%
Phase B LGBM + LR + Similarity	150,028 features	94.43%	96.06%
Phase 3 LGBM + Shallow LGBM	150,028 features	94.46%	96.06%

3) achieved the same recorded public leaderboard accuracy of 96.06%. Their leak-free CV accuracies differed by only 0.03 percentage points (94.46% versus 94.43%), so the experiments do not provide a basis for treating one as meaningfully superior. Furthermore, neither CV nor public LB performance proves that the models rely exclusively on general linguistic patterns rather than dataset-specific regularities.

Controlled Capacity Ablation. The full-capacity LightGBM reaches 100% training accuracy. High training accuracy alone does not establish harmful overfitting; the relevant evidence comes from performance on held-out folds and independent leaderboard evaluation. Under the same reduced feature representation (20,000 TF-IDF features, Heaps’ Law, and Graph features), reducing model capacity did not improve validation accuracy. The full-capacity LightGBM achieved 94.28% validation accuracy, compared with 93.93% for a regularized LightGBM, 93.55% for a low-capacity LightGBM, and 91.84% for a low-capacity XGBoost model. This experiment does not prove the absence of memorization or leakage; it only shows that the tested capacity reductions did not improve validation performance under this setup.

Limitations include a restricted hypothesis space and the fact that public leaderboard feedback inherently creates model-selection bias risk. The empirical 94.38% hurdle is a pragmatic project-specific threshold, not a formal measure of statistical significance. The token mapping is unavailable, limiting semantic interpretation, and the fixed dataset limits claims about transfer to other sources or unseen generation systems. Finally, the 150k-dimensional sparse representation increases computational requirements.

6 Conclusion

This project engineered a 150,027-dimensional representation, and subsequently a 150,028-feature similarity-augmented representation, to classify obfuscated token sequences. Two final ensemble configurations (Phase B and Phase 3) achieved the same recorded public leaderboard accuracy of 96.06%. The small 0.03 percentage-point difference in cross-validation accuracy between them (94.46% vs 94.43%) does not establish that one is universally superior. More importantly, correcting early data leakage led to a fold-isolated, multi-seed evaluation protocol. This framework provided an empirical criterion for assessing features and demonstrated that higher local validation performance does not necessarily translate into higher public leaderboard performance. Capacity ablation showed that reducing capacity did not improve validation performance in the tested reduced-feature setup.

AI Usage

AI tools were used during the project to assist with code development, debugging, feature-engineering exploration, experiment interpretation, and report drafting. The experimental results reported here are based on the project’s recorded experiment outputs.

References

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